AMI Meter
Residential

E350

U3400
RELEASE NOTES FOR F/W VERSION R4.5

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Version	Date	Comments
1.0	11/02/2016	Initial Version

Important Information

This document provides information relating to E350 U3400 meter firmware (the "Product")

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1. About This Document

1.1. Purpose

The purpose of the Release Notes document is to communicate major new features and changes in this release of U3400 firmware. It also documents known problems and workarounds.

1.2. Scope

This document describes the E350 - U3400 firmware version R4.5.

1.3. Definitions, Acronyms, and Abbreviations

LC: Load Control
QoS: Quality of Supply
LP: Load Profile
AMI: Advanced Metering Infrastructure
SCC: Supply Capacity Control
RSM: Relay State Machine
PP: Partial Programming

1.4. References

E350 U3400 User Manual

2. About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released Firmware	This Released	Compatible Meter	Upgradability	Downgradability
R4.3	**R4.5**	U3400	4.3 →4.5	4.5 →4.3

Broadly, this R4.5 firmware release provides following information:

(i) Gross Metering or Net Metering
(ii) kVA demand registration
(iii) Meter display for SCC timer
(iv) Configurable Breaker arming timeout
(v) Other F/W improvements as detailed in the document
(vi) Limitation of firmware

This firmware has been tested on the following platforms

- E350 U34xx meter
- EMPWin+ 2.8

3. New Features, Important Notes and Fixes

3.1. New Feature: Gross or Net Metering

This R4.5 release now allows support for Gross **or** Net (current default state) metering in the kWh Imported Total, kWh Export Total, kVARh Total (Q1+Q2), kVARh Total (Q3+Q4) registers and kVAh Total registers. All previous F/W versions automatically performed Net metering so that only the net difference between import and export loads was reported in the Total register as either import or export whichever was the greater at that time. The new Gross metering option allows the import and export loads to be recorded independently of each other in the two Total registers. The meter needs to be fully programmed to change from one mode to the other and this will automatically **clear all previously recorded LP or Billing data**.

The Metering Type Gross or Net is now reported in the Program read, LP read and Billing read reports if read with **EMPWin+ 2.8 or later.**

The Power Active Total, Power Reactive Total and Power Apparent Total displayed and reported in a Billing read will always be a Net value regardless of the Gross or Net setting. The individual Instantaneous Power measurements combined with the Element Power Factors can be used during installation to confirm the magnitude and the direction of the element loads.

Note that:

- If Gross metering is being used as the metering method then any Total registers being used in LP will have Gross values. If metering is Net then the Total registers will be Net values.
- Full programming will be required to change the metering method from Net to Gross or back to Net again.
- Full programming with older programs that didn't support the Gross or Net metering option will result in the meter recording Net values.

3.2. New Feature: kVA/kVAh Metering

This R4.5 F/W now supports kVA and kVAh metering. The kVAh is calculated using the existing current and voltage registers in the meter. Table 1 provides details of the kVAh Energy registers.

Table 1 below provides available kVAh (energy) registers for use in TOU register selection, LP channel selection and Pulsed Outputs selection.

Table 2 below provides the available kVA (demand) registers for TOU register selection.

Additionally:

- New display lists have been created for all eight new energy registers and four new Apparent Power display registers.
- Billing reads will now automatically include the eight new kVAh total registers.
- For the TOU Display options, items for kVAh support or kVA Demand will automatically be made available if they have been selected as TOU energy or Demand registers.
- This new F/W will allow any of the eight new kVAh registers to be selected as pulsed output channels. The pulsed outputs for any of the Total registers will always be Net even if Gross metering has been selected.

Table 1: U3400 kVAh Energy Registers

Energy Type	Measuring Element(s)	Metering Quadrant
kVAh	kVAh Phase A	Q1+Q4, Q2+Q3
	kVAh Phase B	Q1+Q4, Q2+Q3
	kVAh Phase C	Q1+Q4, Q2+Q3
	Total	Q1+Q4, Q2+Q3

Table 2: U3400 kVA Demand Registers

Demand Type	Measuring Element(s)	Metering Quadrant
kVA	kVA Phase A	Q1+Q4, Q2+Q3
	kVA Phase B	Q1+Q4, Q2+Q3
	kVA Phase C	Q1+Q4, Q2+Q3
	Total	Q1+Q4, Q2+Q3

3.3. New Feature: SCC Remaining Time on display

There is a new display option that has been made available to support the SCC Re-close Timer countdown value. It is only made available in the Normal Display list. It is shown as either ‘SCC t’ or with the remaining Breaker reclose time in hh.mm.ss format.

When the Breaker has been opened due to a Supply Capacity Control operation the meter will now display how long before the Breaker will reclose. The displayed item is a clock countdown value and when it reaches 00.00.00 the Breaker reclose operation will start.

The SCC t (SCC Remaining Time display) item must be manually selected to be part of the Normal Display list. If it is selected to be part of the Normal Display list, when it is not activated it will be displayed as 'SCC t' in the Normal Display.

If SCC opens the Breaker then the next time the Normal Display reaches the SCC t, the display will lock on this and the remaining time before the Breaker recloses will be continually displayed. Manual scrolling through the other display items is still supported even while the countdown is running.

Note that:

- Partial Programming (PP) of the new display item SCC Timeout is supported by the normal Partial Programming of Display items.

3.4. New Feature: Breaker ARMing Timeout

There has been an option added in this F/W release to allow the Breaker ARMed state to be automatically terminated if the Alt2 button has not been used to complete the Breaker operation within a specified time period. The Breaker Arming Timeout value is in the Relay Settings page of MeterDev. If 0 is selected then Breaker Arming operation will be the same as in all previous F/W builds which means the Alt2 button must be pressed to complete the Breaker operation. The timeout can be set in hours from 0 to 48. If a power outage occurs during the Arming timeout period then on power up the remaining time will be reduced by the outage duration period (provided the outage was for at least one minute). If the ARMing timeout period expires then an event will be logged and the meter display returns to the Normal Display list in the meter. **The Breaker will remain in the same state as it was prior to the ARMing command being sent to the meter.**

Sending the same Breaker operation command while in the ARMed state, will cause the ARMing timeout period to restart again. If the opposite Breaker command is sent then ARMed state is terminated without the Breaker switching. If the Breaker ARMing timeout value is Partially Programmed then any currently active ARMed Breaker action will be terminated without the Breaker switching.

In previous F/W, it was not possible to disable the ARMed state once the ARMing command was sent.

This feature works in both Credit and Prepayment mode.

Note that:

- Partial Programming (PP) of the Breaker ARMing timeout value is supported.

3.5. Important Note: Firmware Upgrade

U3400 can be upgrade from R4.3 to R4.5 via the optical port but not via the remote port using PSEMX.

If a meter already has kWh and kVARh consumption and it is F/W upgraded to R4.5 then the new kVAh registers will automatically start accumulating kVAh registers starting from 0, so they will not be consistent with the kWh or kVARh registers. If the meter is fully programmed then all registers will start again from 0.

3.6. Important Note: Firmware Downgrade (Rollback)

U3400 can be downgraded from R4.5 to R4.3.

3.7. Improvement: Ripple Program Writing:

Repeated cycles of writing a Ripple program and then a non- ripple program will no longer cause the FailSafe operation to not trigger in a valid ripple program.

3.8. Improvement:

The issue that 'the armed state breaker couldn't be re-closed automatically after midnight if the maintenance hour was 00:00' is fixed by making changes in EMPWin+ 2.8. EMPWin+ (MeterDev) now does not allow the user to set exactly 00:00 as the maintenance hour setting in the meter program.

3.9. Improvement:

LP Group 1 will now be selected by default and the user cannot disable this group. This is to avoid any possibility of Invalid LP Partial Programming on U3400 meters.

3.10. Import Note: Prepayment Option

The Prepayment mode supports all of the new Breaker Related R4.5 F/W features.

3.11. Limitations:

- *During lab testing, It has been observed that if a meter is being fully programmed with a date in the future then it is possible that the full programming session might fail with a timeout due to zero filling of the LP data from the meter's current date to the new future date. This is most likely to occur if the meter has the maximum number of LP channels and groups enabled, prior to the full programming. To ensure this doesn't happen during bench testing the LP in the meter can be stopped by performing a time change backwards of about an hour prior to the full programming process.*

- *Partial programming doesn't log the reference program used; this can be tracked via the software tool used if required.*

- *Programming log doesn't indicate the User ID, it is always 777.*

- *Group Broadcast Table write commands that are received by a meter with a non-matching address incorrectly counts this as a Table Write error (it should just ignore the command).*

- *If two Breaker close commands are sent before the Breaker has had time to close then the AD will correctly operate but report 0's for the AD parameters.*

- *There could be a small current value displayed/reported even when no current is flowing in an element. The energy will not be creeping due to this minor problem.*

- *The display item Time Zone is incorrectly displayed if a negative time zone area has been selected e.g. USA time zones. The meter uses the correct TZ offset.*

- *The Alarm icon (bell) that is shown on the Display Test is not currently used.*

- *If the meter is fully programmed while the ARMing state timer is running, the ARMing state is correctly cancelled but an event is not created. This is seen as a minor issue because full programming is often followed by a manual request to clear the event log. Partial programming of the ARMing timers does correctly log an event if the timer is terminated.*

- *If outages are less than about 5 seconds duration then there is a possibility that the LC LOS events will be reported twice in the event log. Once before the actual Power Up event and once after this event. This is not a serious problem just duplicate event logging.*

- *The event associated with Breaker LOS operation (2126) is only being logged when the meter is powered down. On power up, if the Breaker closes due to LOS timeout then it is expected that an additional event will be logged. However it doesn't get logged. This is not a serious problem just a missing event according to the technical specification.*

- *The triggered status of UFLC is not reset by the Toolbox command "Disable UFLC". UFLC is correctly disabled but the existing triggered state is not cleared. This will prevent UFLC from running if it is enabled again.*

- *If the Ripple Power Restore is set to 0 then after a power up the Ripple Time Table settings option will not be used to control the relays. The work around is to not use 0 for this random delay parameter.*

- *The RC icon on the display will slowly flash if the meter contains a valid ripple program and it is in Failsafe mode. If the meter is reprogrammed to disable Ripple via the RSM settings then the slow flashing RC icon is expected to stop but it doesn't. It requires a power cycle to terminate the slow flashing of the RC icon.*

- *If the reconnect period is short and the Breaker reconnects within the same Trading period then the Breaker will disconnect at the next minute boundary, therefore short reconnect periods are not recommended.*